Module 2 — Synthesis Quiz

Question bank

Module 2, Synthesis quiz

Readings: Read 0, Read 1, Read 2
Labs: Demo 1 — tools, Demo 2 — worked, Exercise

Section A — Interpret confidence intervals

Use: *"I am 95% confident that [parameter] in the entire population is between [L] and [U]. In this context, confidence means that if data from many samples were collected the same way, then 95% of the intervals computed from these samples would contain the true parameter."*

A1. First-generation proportion

95% CI for proportion of undergraduates who are first-generation: [0.22, 0.30]. Interpret in context.

A2. Website minutes mean

95% CI for mean minutes/day on course websites: [41, 55] minutes. Interpret.

A3. Schools older than 50 years

95% CI with FPC for proportion of schools older than 50 years: [0.14, 0.27]. Interpret.

A4. Spot incorrect interpretation

Which is the best interpretation of a 95% CI?

A) 95% probability the parameter is in this interval.

B) Repeated sampling: ~95% of such intervals contain the parameter.

C) 95% of individuals fall between the endpoints.

Section B — Demo 1 tools (R output & choices)

B1. binom vs prop

$n = 16$, $x = 3$ deaths (mortality proportion). Which function for a 95% CI and why?

B2. prop.test output

`prop.test(x=332, n=400, correct=FALSE)` (seatbelt use). 1. Parameter? 2. Approximate 95% CI? 3. Large-count OK?

B3. t.test output

`t.test(c(8,10,12,12,14,16))` for library hours. Parameter? FPC needed if $N = 120$ and SRS without replacement?

B4. Normal tools analogy

For numerical outcomes, which pair is the normal analogue of `dbinom` / `pbinom` for counts?

B5. Dot plot vs histogram

You have $n = 6$ library hour values. Which ggplot layer shows each duplicate clearly?

B6. survey FPC

In `svydesign(..., fpc = ~N)`, what does the `fpc` argument tell R?

B7. prop.test() decision

When using `prop.test(x, n, p = p0)` for a hypothesis test, which output do you compare to α?

Section C — Tidyverse summaries (Read 2 · Project 2 graphs)

These dplyr verbs support mean/median summaries and graphs in Read 2 and Project 2.

C1. Pipe purpose

Main purpose of `x %>% f()` in tidyverse?

C2. filter()

Which function keeps rows satisfying a condition like `hwy > 30`?

C3. select()

Which function chooses columns by name?

C4. summarize()

Which function computes summary statistics (often fewer rows)?

C5. mutate()

Which function adds a new column with the same number of rows?

C6. count()

Which function counts rows per category?

C7. group_by()

Role of `group_by()` before `summarize()`?

C8. n()

Inside `summarize()`, what does `n()` return?

C9. glimpse()

Name two things `glimpse(df)` shows about a dataset.

Section D — Type I/II errors in context

D1. Lucky coin

Test if coin favors heads ($p = \mathrm{P}(\text{heads})$). State H_0, H_a, Type I error, Type II error in context.

D2. Lucky die

Test if die favors 6. State H_0, H_a, Type I and Type II errors in context.

D3. Medical treatment

Test if treatment effective in more than 10% of cases. State hypotheses and both error types.

D4. Left-handed students

Test if school left-handed rate exceeds 10%. State hypotheses and both error types.

D5. Defective supplier

Test if defect rate below 15%. State hypotheses and both error types.

Section E — Decisions & possible errors

Use $\alpha = 0.05$ for each problem.

E1. 62 heads in 100

$H_0 : p = 0.5$ vs $H_a : p > 0.5$; 62 heads; p-value = 0.0096.
Reject? Type I possible? Type II possible? State Type I in context.

E2. 4 left-handed in 25

$H_0 : p = 0.10$ vs $H_a : p > 0.10$; 4/25 left-handed; p-value = 0.21.
Reject? Type I possible? Type II possible? State Type II in context.

E3. 3 defective in 40

$H_0 : p = 0.15$ vs $H_a : p < 0.15$; 3/40 defective; p-value = 0.034.
Reject? Type I possible? Type II possible? State Type I in context.

Section F — Reducing errors

F1. Reduce Type I

Which helps reduce Type I error?

A) Larger α B) Smaller α C) Increase n D) Decrease n

F2. Reduce Type II (fixed α)

With fixed α, what increases power / reduces Type II error?

A) Smaller α B) Larger α C) Increase n D) Decrease n

Section G — Demo 2 worked examples (Read 2 workbook)

Match each Demo 2 / Read 2 example to the correct interval family, conditions, and FPC decision.

G1. Example 1 — seatbelts

SRS $n = 400$, $x = 332$ seatbelt users; statewide population.

Parameter? Method? FPC?

G2. Example 2 — St. Joe's mortality

$n = 16$, $x = 3$ deaths.

Which method? Why not Wald?

G3. Example 3 — plant-based entrée

$N = 80$, $n = 20$, $x = 6$ yes.

Parameter? FPC? Borderline Wald note?

G4. Example 4 — library hours

$N = 120$, $n = 6$ hours: 8,10,12,12,14,16.

Parameter? FPC? t vs z?

G5. Example 5 — commute times

$n = 40$, $\bar{x} = 28.5$ min, $s = 8.2$; large metro.

Parameter? FPC? CLT role?

Section H — Exercise lab (new stories)

Parallel to exercise lab — application and interpretation.

H1. Flu vaccine proportion

$n = 350$, $x = 266$ vaccinated; large state population.

Method? One-sentence CI interpretation template?

H2. Drug side effect

$n = 14$, $x = 2$ with side effect.

Check conditions. Preferred CI function?

H3. Dorm study lounge

$N = 95$, $n = 22$, $x = 8$ with lounge.

FPC factor formula? Why multiply SE?

H4. Espresso temperature mean

$N = 60$ shops, $n = 12$ temperatures (°F). Which plot first? Which CI family?

H5. Daily steps mean

$n = 35$ step counts; large app cohort.

FPC? Conditions checklist (name two)?

Section I — Read 1–2 application & conditions

I1. Test vs CI (proportion)

You want a 95% CI for p only (no p_0). Do you pass `p =` to `prop.test`?

I2. Wald by hand

$\hat{p} = 0.83$, $n = 400$. Compute approximate SE and MOE (multiplier 2).

I3. Plus Four when

When is Plus Four a reasonable alternative to exact binomial for a proportion CI?

I4. Population vs sample

$\bar{x} = 12$ library hours from $n = 6$ libraries. Is 12 a parameter or statistic?

I5. FPC threshold

When should you consider applying the FPC in Project 2?

I6. Wrong CI language

Student writes: 'There is a 95% chance the true mean is in [25.9, 31.1].' What is wrong?
